

UPS Product Tracking Information System

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COURSE NAME: Database Management Systems

COURSE CODE: CSA0503

COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Short Answer Text(High)

Assessment Tool Weightage: 30%

Question 1: UPS Product Tracking Information System

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.

UPS Product Tracking Information System – Entity Relationship (ER) Diagram

Entities and Attributes

1. SHIPPED_ITEM

Primary Key: ItemNumber

Attributes:

- ItemNumber (PK)
- Weight
- Dimensions
- InsuranceAmount
- Destination

- FinalDeliveryDate

2. RETAIL_CENTER

Primary Key: CenterID

Attributes:

- CenterID (PK)
- Type
- Address

3. TRANSPORTATION_EVENT

Primary Key: ScheduleNumber

Attributes:

- ScheduleNumber (PK)
- Type (Flight/Truck)
- DeliveryRoute

Relationships

RECEIVED_AT

Relationship between

RETAIL_CENTER ←—— Receives ——→ **SHIPPED_ITEM**

Cardinality

One Retail Center receives many shipped items.

One shipped item is received at exactly one retail center.

Retail Center (1) -----< Receives >----- (M) Shipped Item

TRANSPORTED_BY

Relationship between

SHIPPED_ITEM ←—— Transported By ——→ **TRANSPORTATION_EVENT**

Cardinality

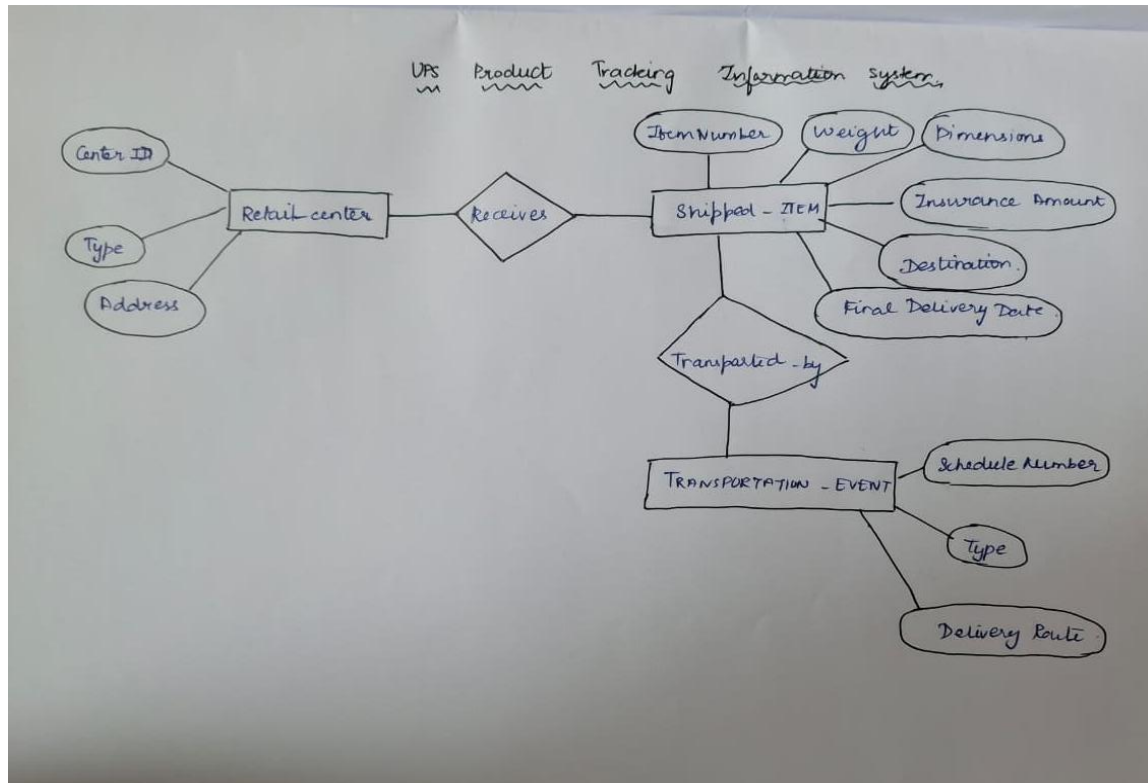
A shipped item travels through one or more transportation events.

A transportation event can transport many shipped items.

This is a **Many-to-Many (M:N)** relationship.

Shipped Item (M) -----< Transported_By >----- (M) Transportation Event

ER Diagram:



Cardinality Summary

Relationship	Cardinality
Retail Center → Shipped Item	1 : M
Shipped Item → Transportation Event	M : N

Primary Keys

Entity	Primary Key
Shipped_Item	ItemNumber
Retail_Center	CenterID
Transportation_Event	ScheduleNumber